Probabilistic Heat Stress Forecasting

over Global Coastal Cities

ERA5 Reanalysis Data (2010-2025) | Probabilistic GRU | Uncertainty Quantification

Background & Motivation

- Climate change intensifies **urban heat stress**
- Coastal cities face compound heat-humidity risks
- Traditional NWP models: computationally expensive
- Need: **fast, uncertainty-aware** local forecasts
- Key gap: most ML approaches are **deterministic** — no confidence intervals

Research Questions

1. Can a lightweight GRU produce **calibrated probabilistic** heat stress forecasts?
2. How do predictions vary across **climate zones**?
3. Which architectural components matter most?

Study Sites

Four coastal cities spanning distinct climate zones

City	Climate	Lat/Lon	Characteristics
Qingdao	Temperate monsoon	36.25N, 120.50E	Humid summers, strong seasonality
Dubai	Arid subtropical	25.25N, 55.25E	Extreme dry heat
Singapore	Tropical	1.25N, 103.75E	Year-round high humidity
Miami	Subtropical	25.75N, 80.25W	Hurricane-prone coast

Why these cities? — They represent 4 fundamentally different heat stress profiles, testing model generalization across climate regimes.

Data: ERA5 Reanalysis

16 years of hourly data (2010-2025) from ECMWF ERA5

24 Input Features

- 2m temperature, dewpoint, skin/SST
- 10m & 100m wind components, gusts
- Shortwave & longwave radiation
- Surface & sea-level pressure
- Precipitation
- **Engineered:** RH, wind speed, temp gaps
- **Temporal:** hour & day-of-year (sin/cos)

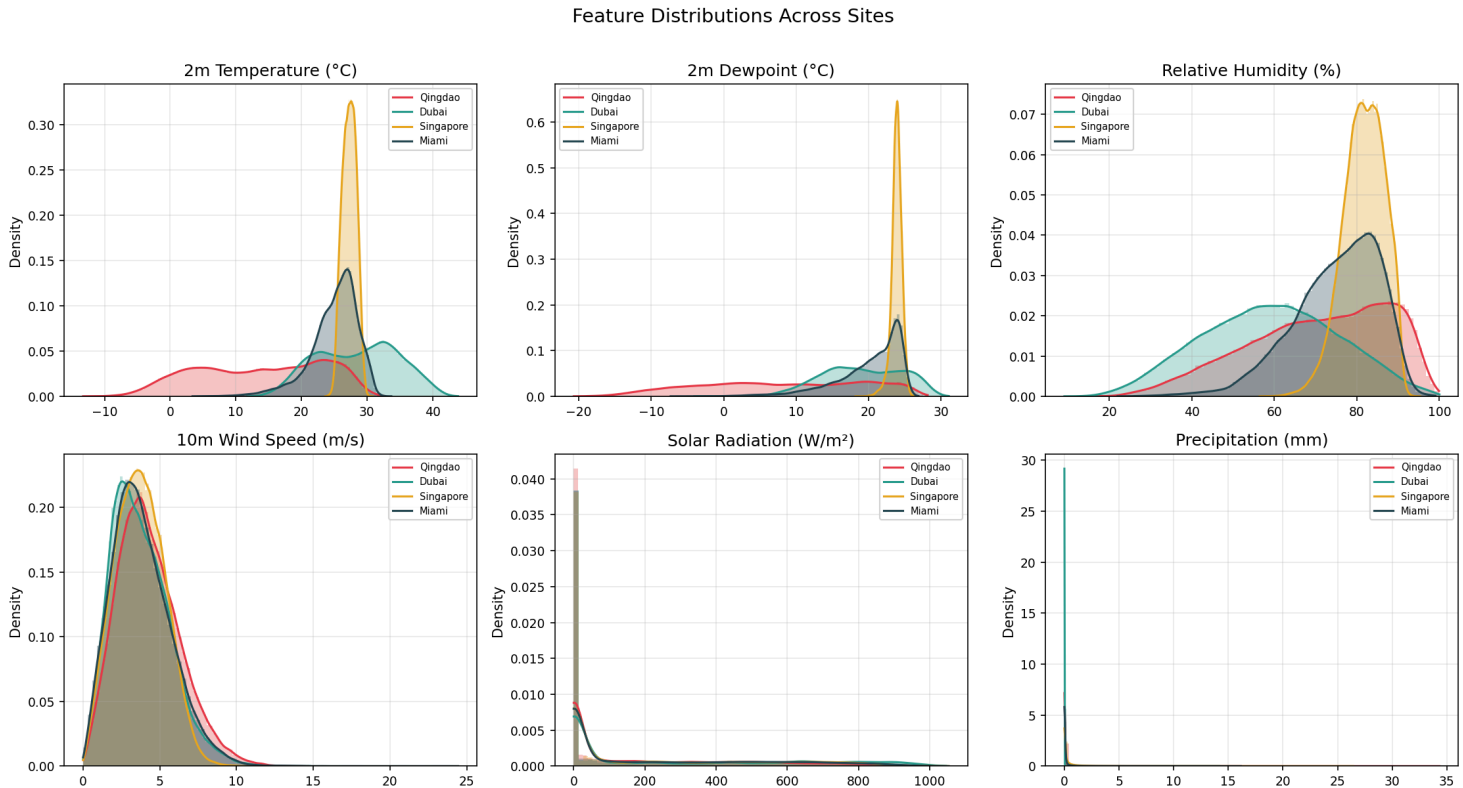
Task Setup

- **Lookback:** 72 hours
- **Forecast horizon:** 24 hours
- **Split:** 70% train / 15% val / 15% test
- **Targets:**
  - Heat Index (HI) — NOAA formula
  - WBGT-like — radiation & wind aware

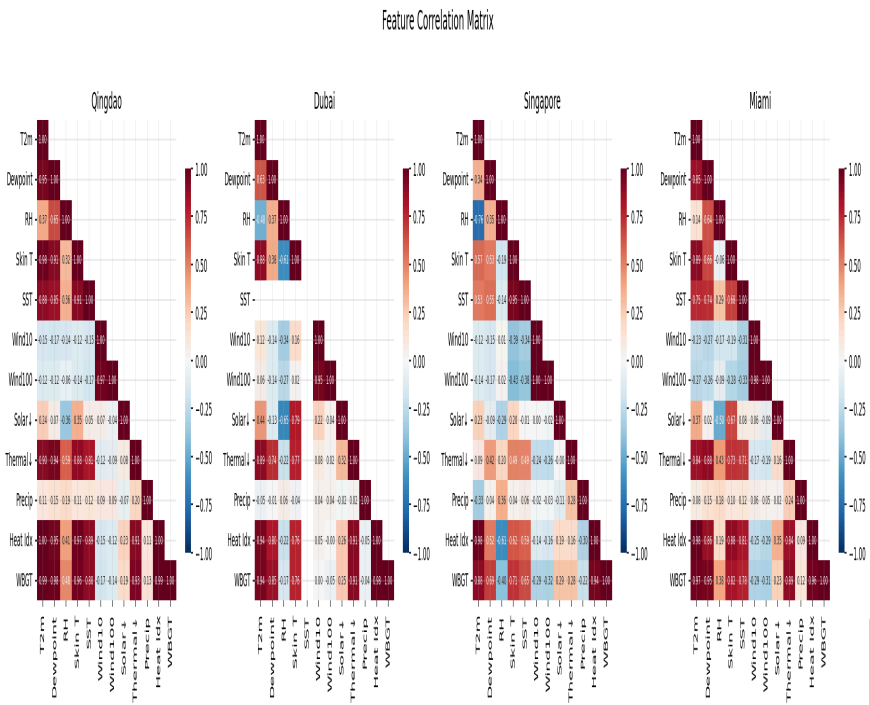
> Temporal split avoids data leakage

Exploratory Data Analysis

Feature Distributions Across Sites



Feature Correlations

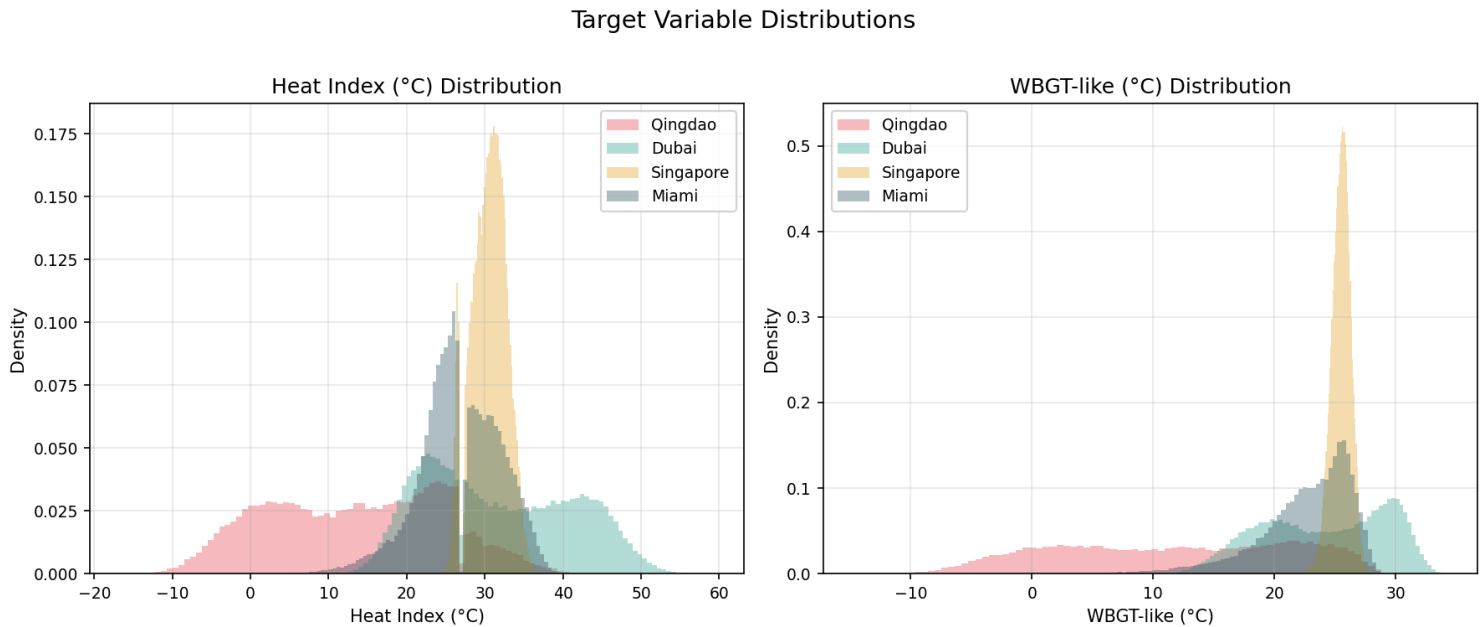


Key Observations

- **t2m ↔ Heat Index**: r = 0.98-1.00 — temperature is the dominant driver
- **Wind10 ↔ Wind100**: r = 0.89-0.96 — redundant but model handles it
- **Solar ↔ Thermal radiation**: negatively correlated in tropics
- Singapore: overall **weaker** inter-feature correlations (narrow temp range)

Correlation structure varies by climate zone — **one model must adapt** to different regimes

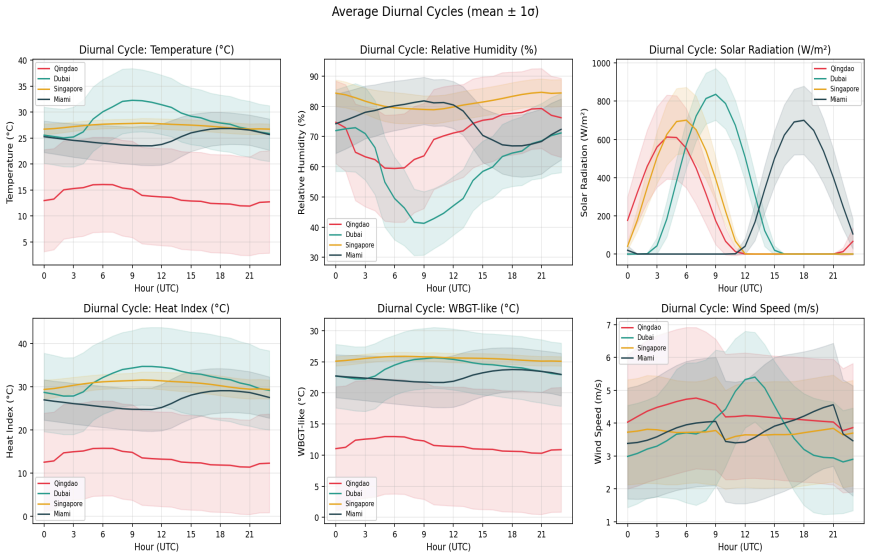
Target Variable Distributions



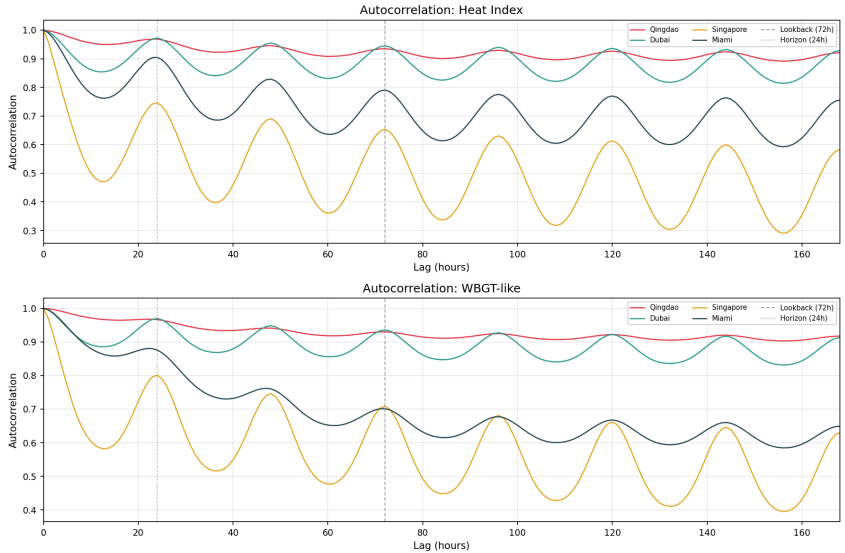
Non-Gaussian targets: Qingdao/Miami are bimodal (winter-summer split), Singapore WBGT is extremely concentrated at 25-28°C. Despite this, our Gaussian output works well because the **conditional distribution** $P(y|x_{1:72})$ is unimodal — the bimodality comes from mixing summer/winter samples.

Temporal Patterns

Diurnal Cycles



Autocorrelation Analysis



72h lookback captures 3 full diurnal cycles — autocorrelation still >0.3 at this lag

EDA Summary

Data Quality ✓

- **No missing values** except Dubai SST (land grid point, handled via skin temperature fallback)
- ~140K hourly samples per site (16 years)
- Stationary after deseasonalization — no long-term drift

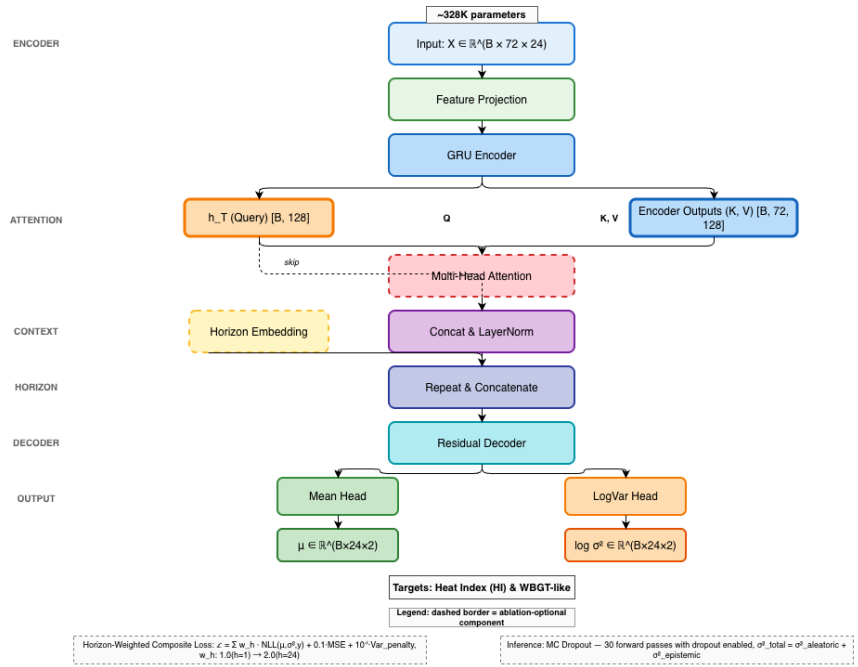
Design Implications

- **Nonlinear model needed**: temperature vs Heat Index is exponential at high-T
- **72h lookback justified**: autocorrelation validates the window choice
- **Gaussian conditional output works**: despite non-Gaussian marginals
- **Climate diversity validated**: 4 sites span extreme ranges in all features

> EDA confirms our architectural and hyperparameter choices are **data-driven, not arbitrary**

Model Architecture

Probabilistic GRU — ~328K trainable parameters



Architecture Details

Encoder

- **Feature Projection**: Linear 24→64 + LayerNorm + GELU
- **Bidirectional GRU**: 2 layers, hidden=128
- **Multi-Head Attention**: 4 heads (optional)

Decoder

- **Horizon Embedding**: Learnable per-step embeddings
- **Residual MLP**: 2 blocks with LayerNorm
- **Dual Output Heads**: μ_t and $\log \sigma_t^2$

Probabilistic Output

$$p(y_t|x) = \mathcal{N}(\mu_t, \sigma_t^2) \quad \text{for each horizon } h = 1, \ldots, 24$$

Epistemic uncertainty via MC Dropout (30 samples at inference)

Training Strategy

Loss Function

Horizon-Weighted Composite Loss:

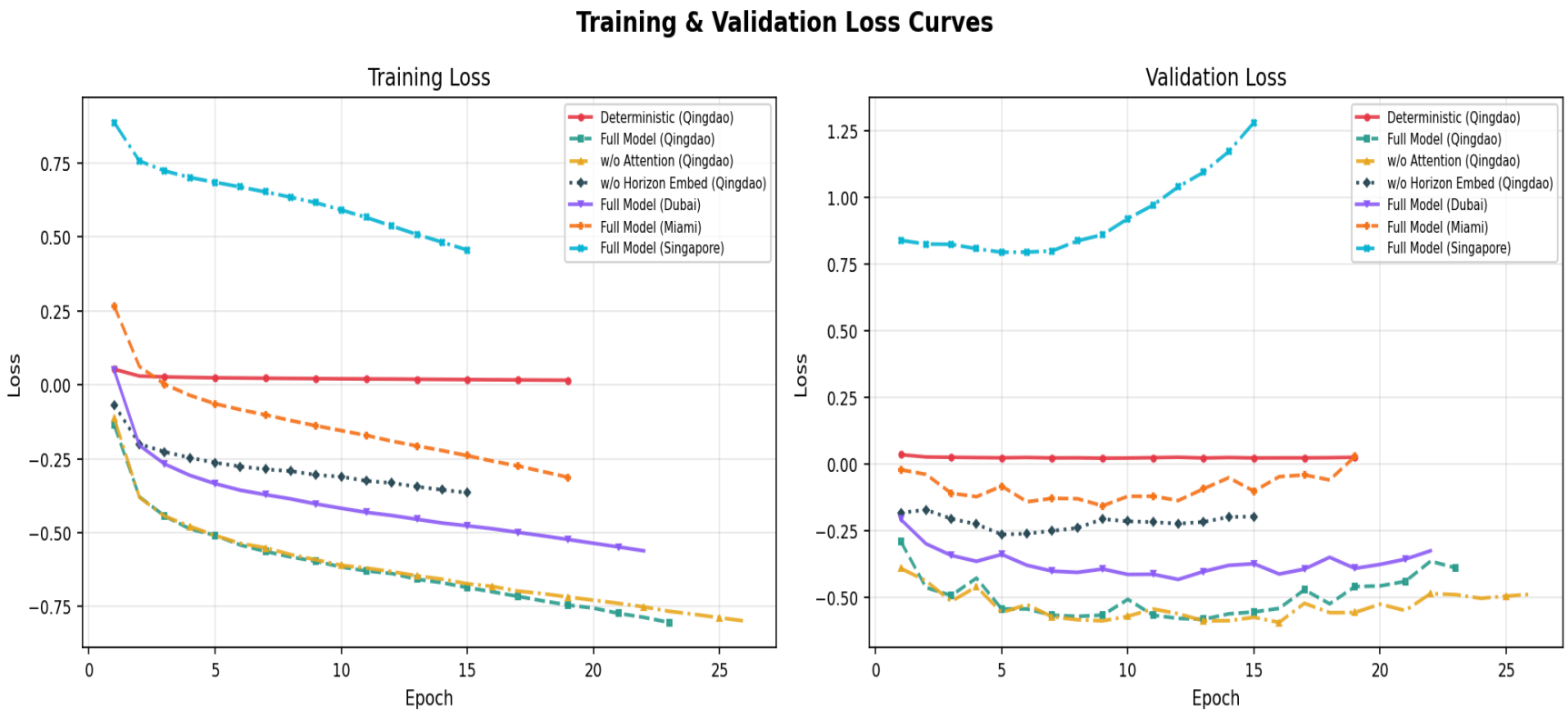
$$\mathcal{L} = \sum_{h=1}^{24} w_h \left[\mathrm{NLL}_h + 0.1 \cdot \mathrm{MSE}_h + 10^{-4} \cdot \sigma_h^2\right]$$

- **NLL**: Gaussian negative log-likelihood
- **MSE**: Stabilizes early training
- **Variance penalty**: Prevents runaway uncertainty
- **Horizon weights**: $w_h = 1.0 \rightarrow 2.0$ (linear)

Optimization

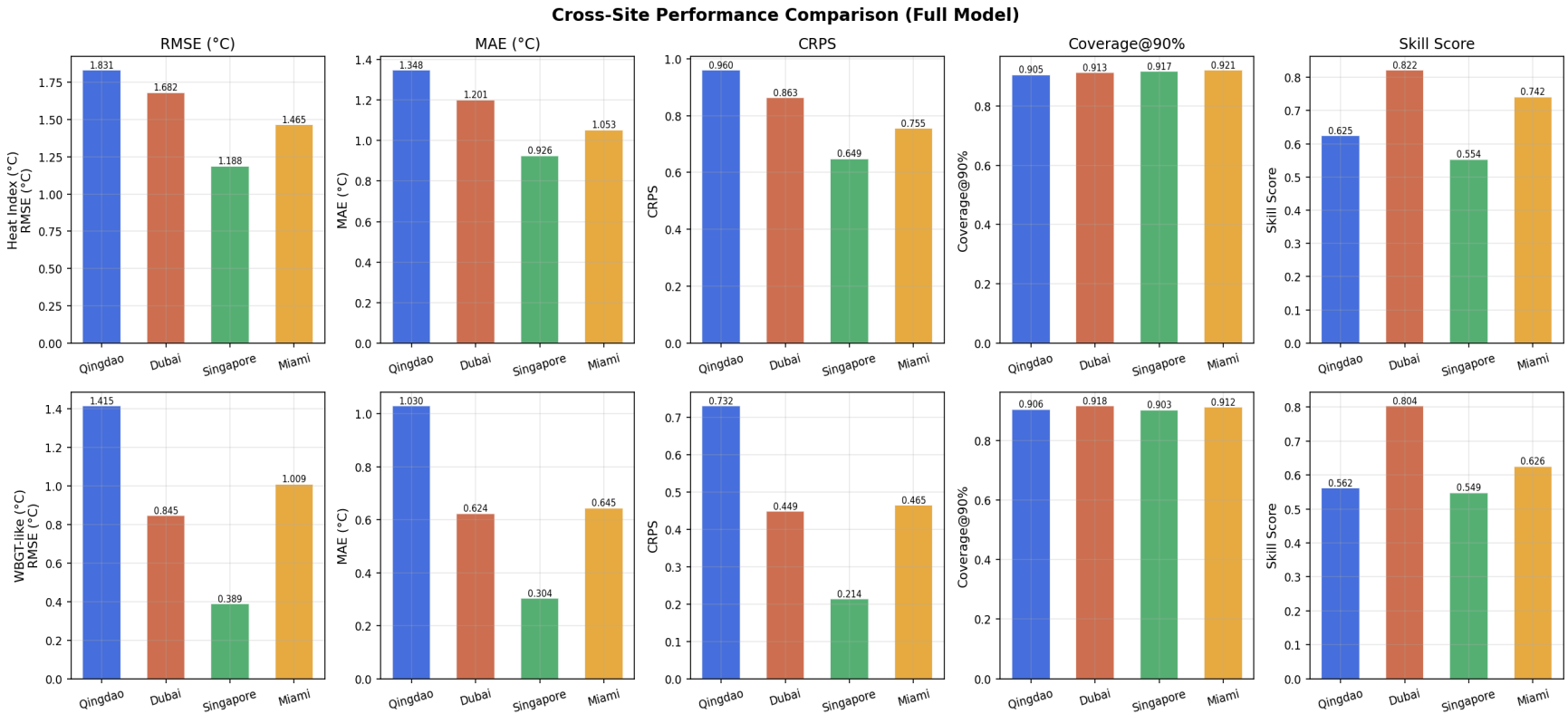
Parameter	Value
Optimizer	AdamW
Learning rate	3e-4
Scheduler	CosineAnnealing
Batch size	256
Max epochs	60
Early stopping	patience=10
Gradient clip	1.0

Training Curves



Results

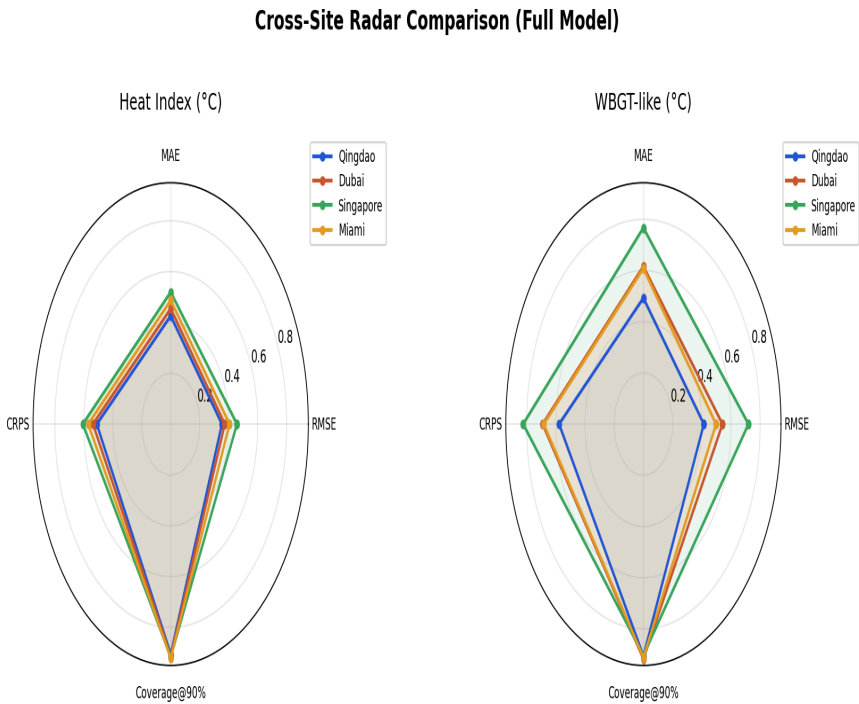
Cross-Site Performance



Key Findings

- **Singapore**: Lowest RMSE (1.19C HI) — tropical regularity
- **Dubai**: Highest Skill Score (0.82) — arid predictability
- **Qingdao**: Highest errors — complex monsoon dynamics
- **Miami**: Balanced performance — subtropical patterns
- **All sites**: Skill Score > 0.55 (beats persistence)

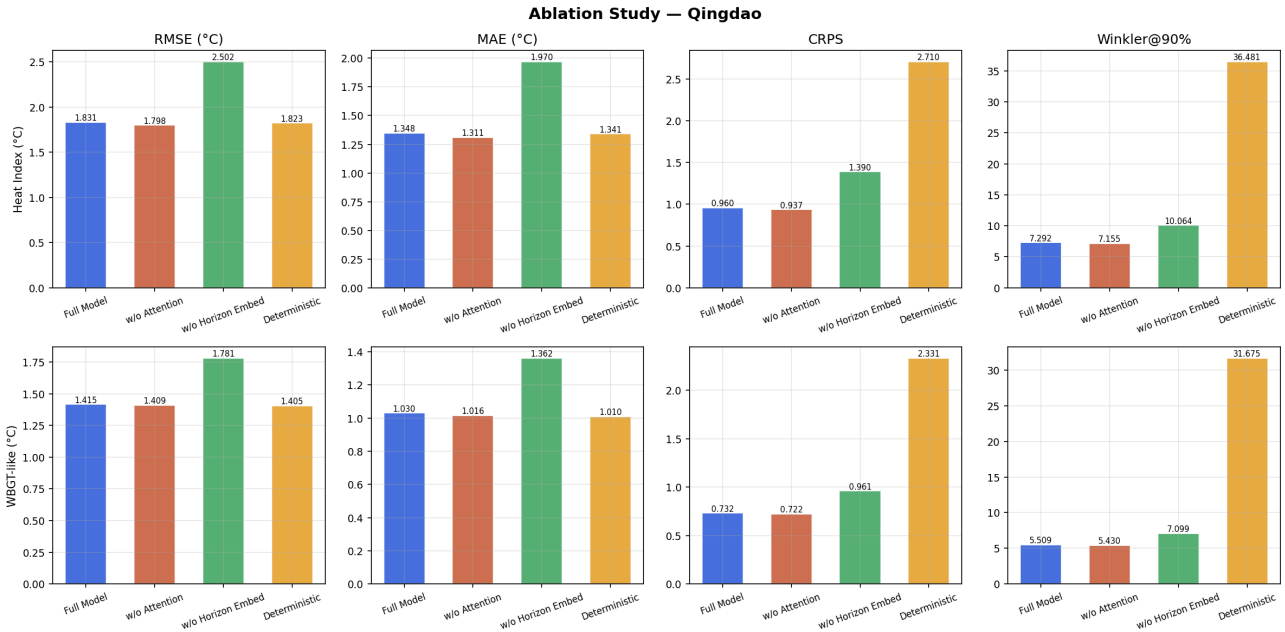
Site	HI RMSE	HI CRPS	Skill
Qingdao	1.83C	0.96	0.63



Ablation Study

Ablation: Which Components Matter?

Tested on **Qingdao** (most challenging site)



Ablation Results

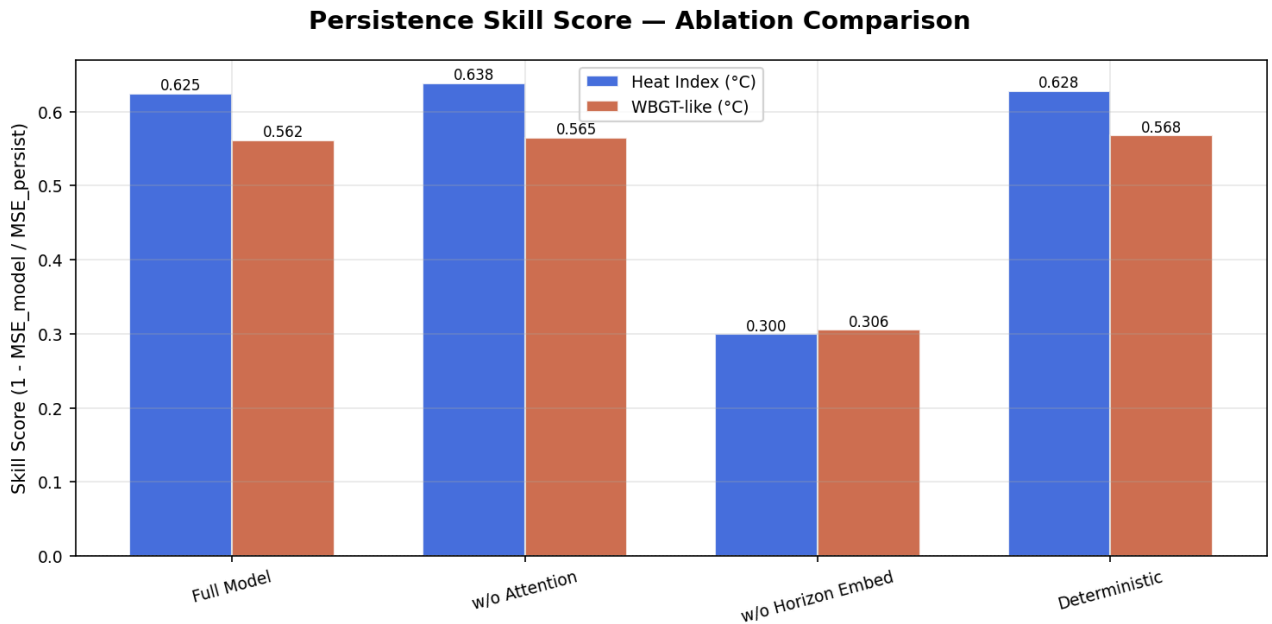
Variant	HI RMSE	HI CRPS
Full Model	1.831	0.960
w/o Attention	1.798	0.937
w/o Horizon	2.502	1.390
Deterministic	1.823	2.710

Three Key Insights

1. **Horizon embedding is critical** — removing it causes **+37% RMSE**
2. **Attention adds no value** — w/o attention is *slightly better* (GRU alone suffices for single-point time series)
3. **Probabilistic output is essential** — deterministic CRPS 2.71 vs 0.96

Takeaway: Invest in output-side design (horizon awareness, probabilistic heads), not encoder complexity.

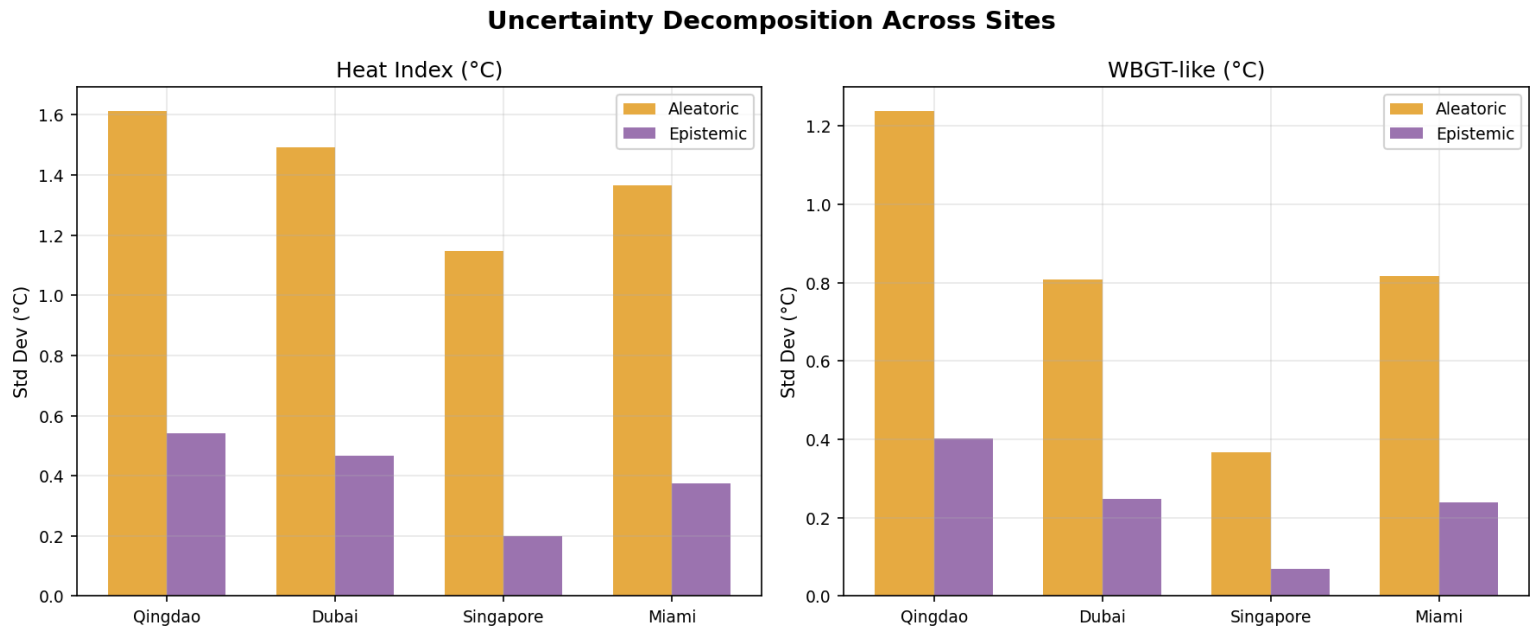
Skill Score vs Persistence



Model advantage **grows with forecast horizon**: ~20% at 1h → ~70% at 24h

Uncertainty Quantification

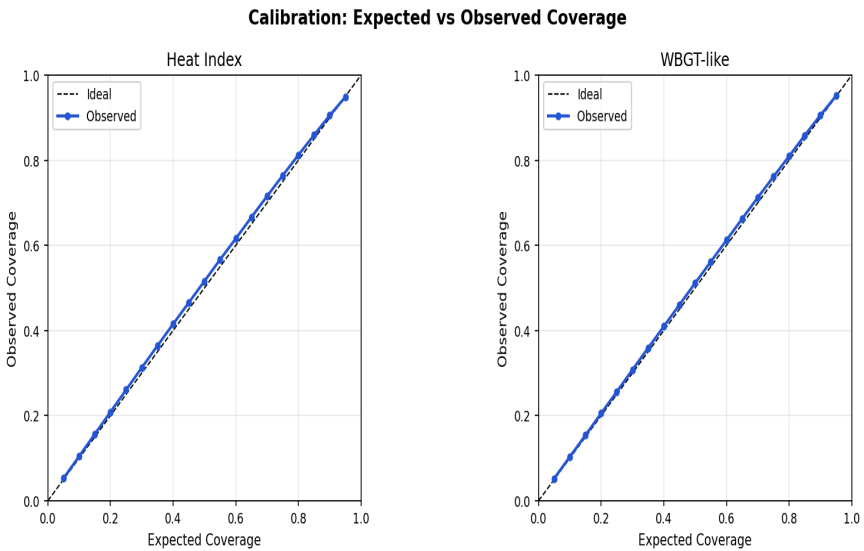
Uncertainty Decomposition



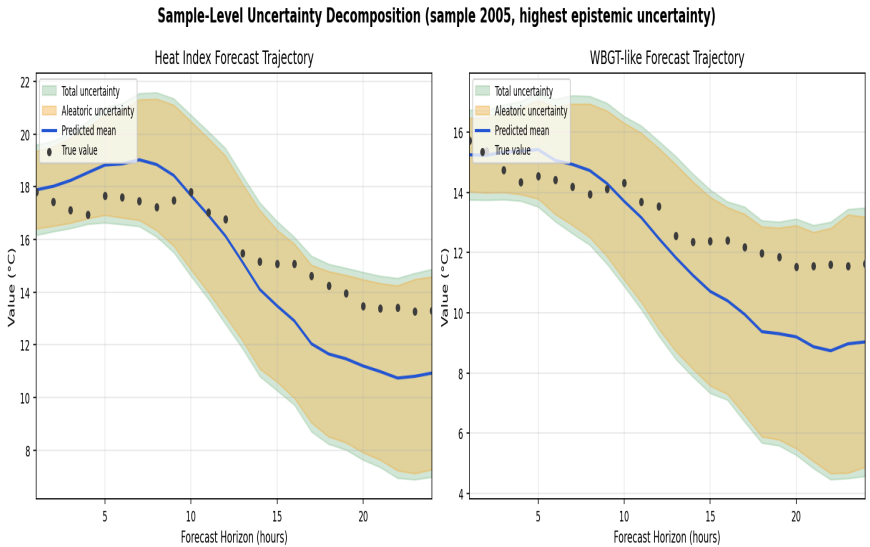
Aleatoric uncertainty dominates (3-5x larger than epistemic) — prediction errors are driven by weather chaos, not model limitations.

Calibration

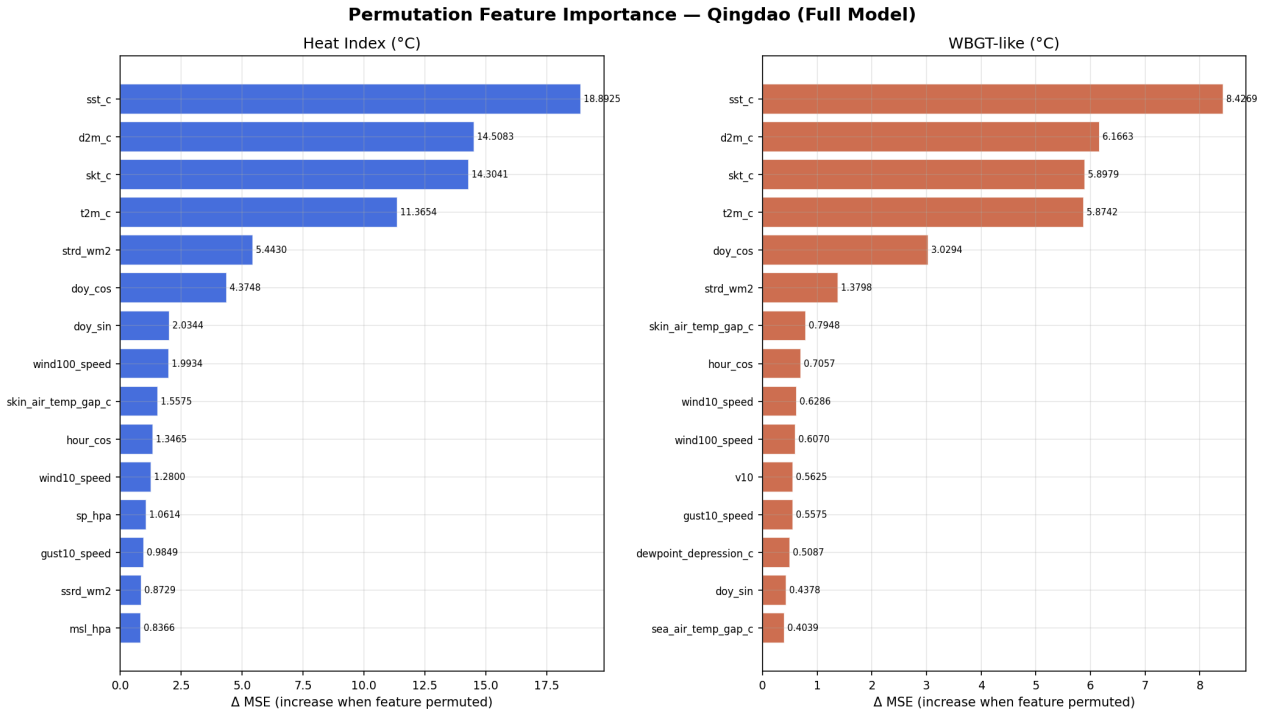
Well-calibrated predictions: 90% intervals cover ~90% of true values



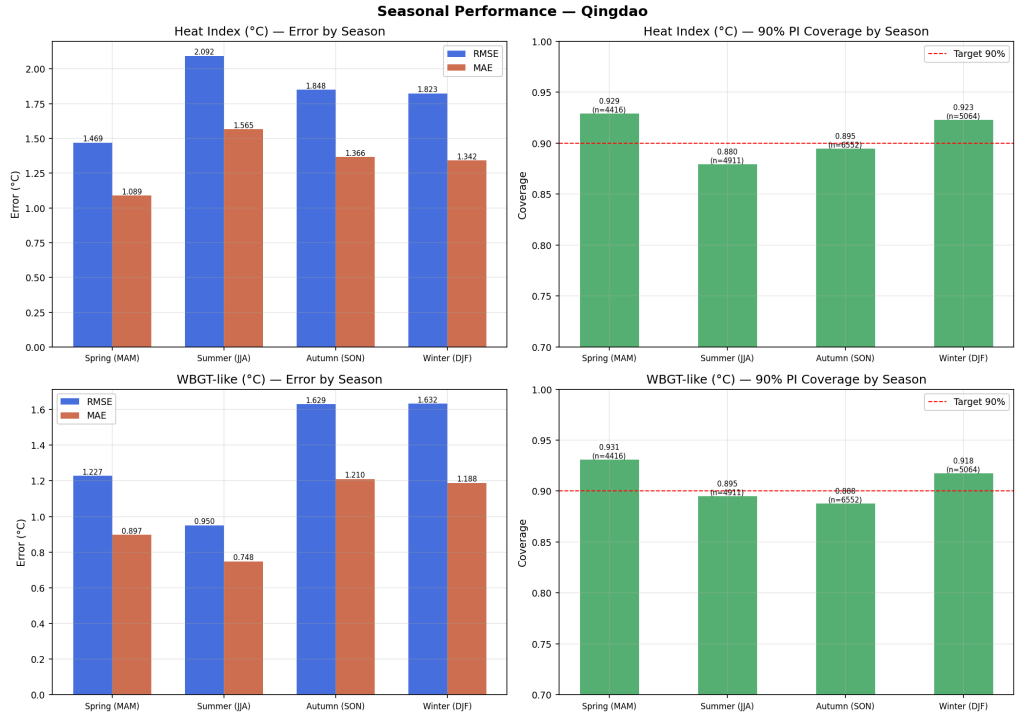
Uncertainty Bands



Feature Importance



Seasonal Analysis



Summary

- **Probabilistic GRU** forecasts heat stress with calibrated uncertainty across 4 climate zones
- RMSE: **1.19-1.83C** for Heat Index
- Skill Scores: **0.55-0.82** (all beat persistence)
- **Horizon embedding** is the most critical component (+37% error without it)
- Attention provides **no benefit** for single-point time series
- **Aleatoric > Epistemic**: weather chaos dominates model uncertainty

Limitations & Future Work

- Single-point ERA5 (no spatial modeling)
- WBGT-like is a proxy, not ISO standard
- 24h horizon only
- Independent site training

Future directions:

- Spatial GNN for multi-point forecasting
- Transfer learning across climate zones
- Conformal prediction for distribution-free intervals
- Integration with NWP ensembles

Thank You

Probabilistic Heat Stress Forecasting over Global Coastal Cities

4 sites | 16 years | ~328K params | Calibrated uncertainty

COMP0197 Applied Deep Learning